a bit clearer

PYTHON AND AI · EPISODE 07

Literals and Variables

How Python Writes, Stores and Changes Data

KEY POINTS

- A **literal** is a fixed value typed into your code: **int** `18`, **float** `3.14`, **bool** `True` / `False`, **str** `"Hello"`, and **None**.
- **Anything in quotes is a string**, even `"10000"`. `True` and `False` need a capital letter, and in maths they count as **1** and **0**.
- A **variable** is a name for a value: `name = value`. The `=` means **assign**: Python works out the right side first, then stores the result on the left.
- A variable holds **one value at a time** and needs a value **before** you use it. `print` shows several values at once, separated by commas.

01 Kinds of Literals

▶ VIDEO 00:33

INT	FLOAT	BOOL	STR	NONETYPE
`10` `18` `50`	`8.5` `3.14` `1.0`	`True` `False`	`"Hi"` `'Hi'`	`None`

Numbers are written as usual; a decimal point makes a **float**. Lowercase `true` is not Python. Without quotes, Python reads text as code.

02 Try Them in PyCharm

▶ VIDEO 02:35

literals.py — PYTHON

```python
# Literals
print(100)  # int
print(3.14)  # float
print(True)  # bool
print(False)  # bool
print("Hello Python")  # str
print(None)  # NoneType

# A bool is a special kind of integer: True counts as 1, False as 0
print(True + 1)
print(False - 1)
```

A value on its own line shows nothing: wrap it in `print`. A `#` after code starts a comment; the code before it still runs.

KEY IDEA

Spot the difference: `"10000"` is a string (text that looks like a number) and `10000` is an integer. `"10000" + 1` fails with a **TypeError**.

03 Create and Change a Variable

▶ VIDEO 05:19

variables.py — PYTHON

```python
# Create a variable: name = value
views = 800
print(views)

# Change it: Python works out the right side first
views = views + 100
print(views)

# Python is dynamically typed: the same name can hold another kind of value
views = "OK"
print(views)
views = True
print(views)
```

PEP 8 recommends a space on each side of `=`. Select several lines and press **Ctrl + /** to comment them all at once. Printing a name you never assigned stops with `NameError: name 'likes' is not defined`.

04 Mini Project: Predict the Views

▶ VIDEO 07:37

views.py — PYTHON

```python
# A video has 3000 views and gets 500 new views every month
base, monthly = 3000, 500

print("Views after one month:", base + monthly)
print("Views after two months:", base + monthly + monthly)
```

`print` separates its items with a space. **Ctrl + D** duplicates the current line in PyCharm. `base, monthly = 3000, 500` assigns the first value to the first name and the second to the second.

CODE 6 files on GitHub github.com/abitclearer/python-and-ai/tree/main/07-literals-and-variables